Indian Institute of Technology, Guwahati
Guwahati, INDIA 781 039

Department of Chemistry

Date: 04 October 2023; 7:55 a.m.

CH101

Tutorial 3; Physical Chemistry

Q1. The hydrogenic Hamiltonian for the electron with mass m_e and nucleus with mass m_N is:

$$H = \hat{E}_{k,electron} + \hat{E}_{k,nucleus} + \hat{V} = -\frac{\hbar^2}{2m_e}\nabla_e^2 - \frac{\hbar^2}{2m_N}\nabla_N^2 - \frac{Ze^2}{4\pi\varepsilon_0 r}$$

Here r is the distance of the electron from the nucleus and ε_0 is the vacuum permittivity. Show that the same can be written as,

$H = -\frac{\hbar^2}{2m}\nabla_{c.m.}^2 - \frac{\hbar^2}{2\mu}\nabla^2 - \frac{Ze^2}{4\pi\varepsilon_0 r}$**. The terms have their usual meanings.**

Q2. Find the mean radius of an electron in the 1s orbital of H-atom. The wavefunction, $\Psi = \frac{1}{(\pi a_0^3)^{1/2}} e^{-r/a_0}$ and $\int_0^\infty r^3 e^{-2r/a_0} dr = \frac{3a_0^4}{8}$ and

$4\pi = \int_0^\pi \int_0^{2\pi} \sin\theta d\theta d\varphi$. Also, $a_0 = 0.529\,\mathring{A}$

Q3. In $H_2C{=}CH_2$, the carbon atoms are sp^2 hybridized. The central carbon atoms have electronic configuration $1s^2 2s^2 2p^2$. Find out the other two hybridized orbitals if one of them is $h_1 = (1/3)^{1/2}\{s + 2^{1/2}p_y\}$. Also, schematically draw the skeleton of one of the carbon atom hybrid orbitals and then show that these orbitals generate a bond angle of 120^o.

Q4. Consider an electron (particle) on a ring. Write the one-electron Hamiltonian for the system. Find the normalized wavefunction (general expression), quantum numbers and energy (generalized expression) of the electron. Use the cyclic boundary condition: $\Psi(\varphi + 2\pi) = \Psi(\varphi)$. **What is the expression for the wavelength of excitation between the first two energy states?**